

Problem 1 (§5, #2, 10, 12). Prove by contrapositive.

- a) Suppose $n \in \mathbb{Z}$. If n^2 is odd, then n is odd.
- b) Suppose $x, y, z \in \mathbb{Z}$ and $x \neq 0$. If $x \nmid yz$, then $x \nmid y$ and $x \nmid z$.
- c) Suppose $a \in \mathbb{Z}$. If a^2 is not divisible by 4, then a is odd.

Problem 2 (§6, #4, 8, 10). Prove by contradiction.

- a) There exist no integers a and b for which $21a + 30b = 1$.
- b) The number $\sqrt{6}$ is irrational.
- c) Suppose $a, b, c \in \mathbb{Z}$. If $a^2 + b^2 = c^2$, then a or b is even.

Problem 3. Prove the following statements using any method. State which method you will be using at the beginning.

- a) The sum of any five consecutive integers is divisible by 5.
- b) (5.18) If $a, b \in \mathbb{Z}$, then $(a + b)^3 \equiv a^3 + b^3 \pmod{3}$.
- c) (5.28) If $n \in \mathbb{Z}$, then $4 \nmid (n^2 - 3)$.